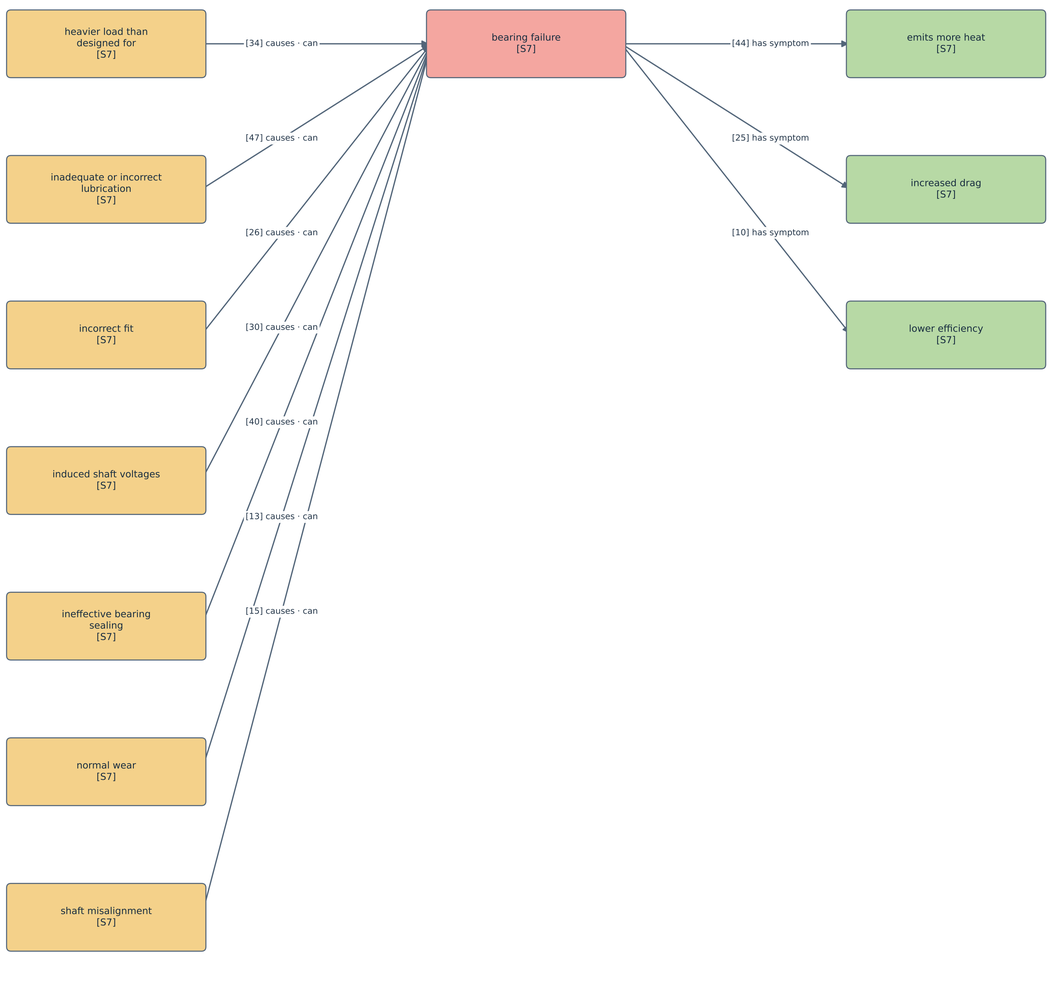
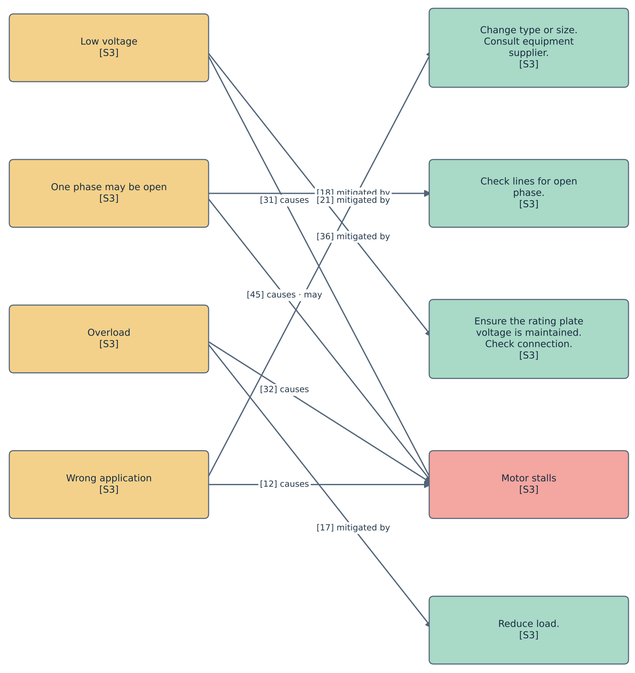
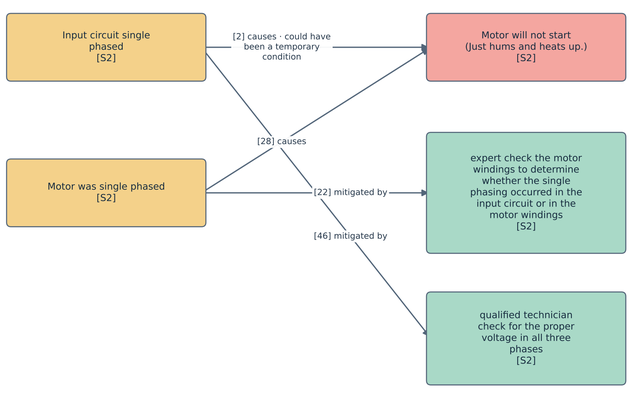
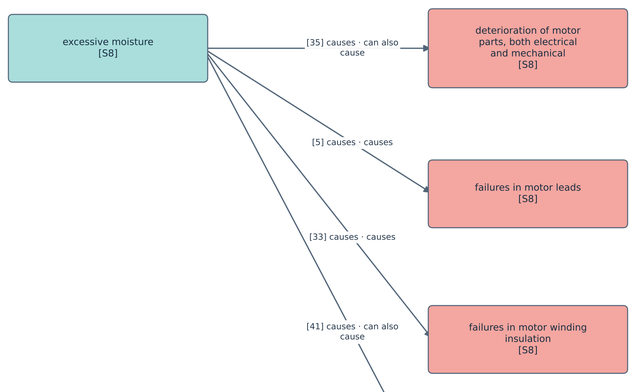
Maintenance knowledge graph

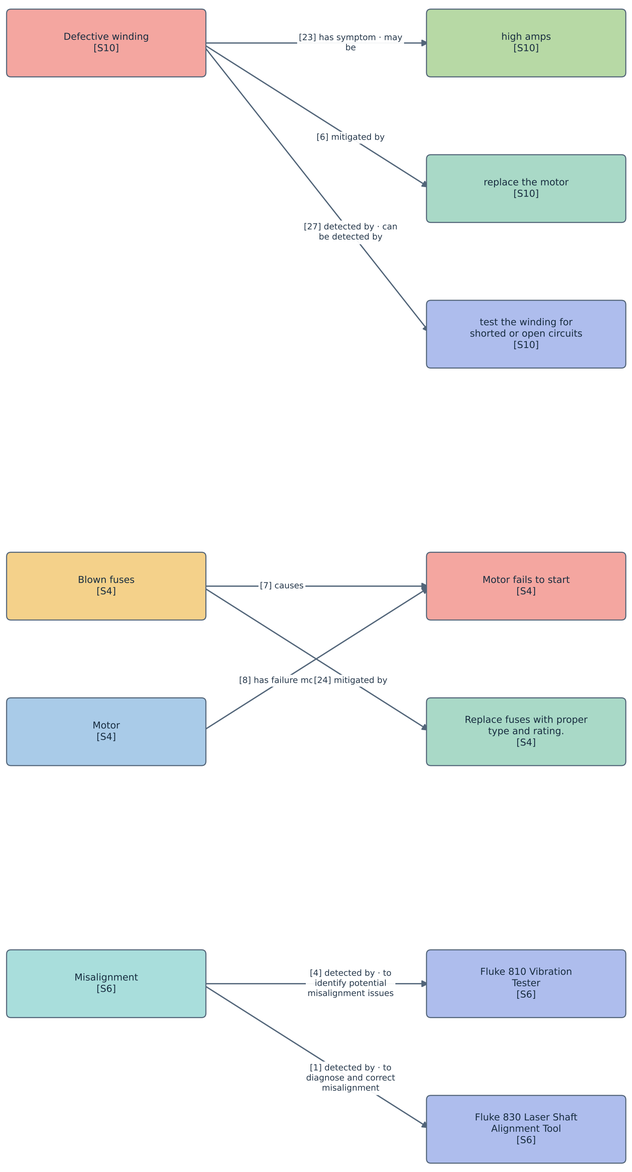
62 concepts · 47 source-supported statements · one graph, with disconnected components preserved



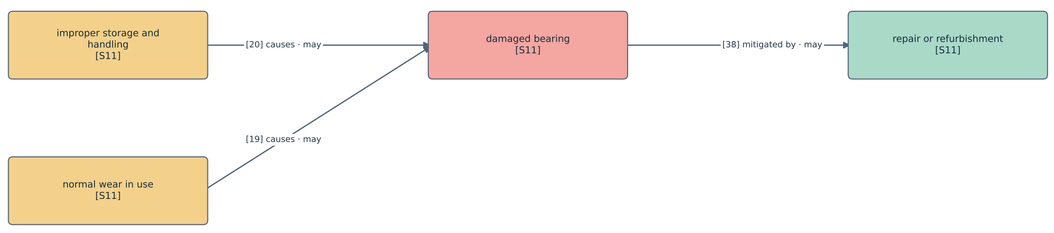


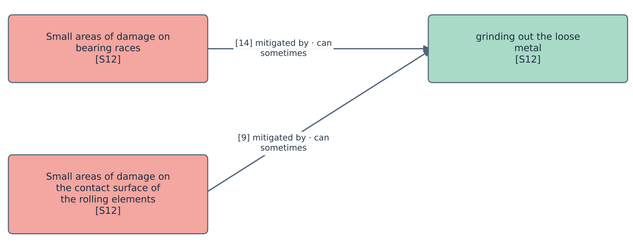






















Failure Mode · Component · Cause · Symptom · Phenomenon · Detection Method · Maintenance Action

Source key — tags on graph nodes

S1: Electric Motor Troubleshooting (Polyphase).pdf — Starting: motor controller

S2: Electric Motor Troubleshooting (Polyphase).pdf — Motor will not start; hums and heats

S3: 3GZF500730-85 RevH_EN 06-2022_lowres.pdf — Motor stalls

S4: 3GZF500730-85 RevH_EN 06-2022_lowres.pdf — Motor fails to start — fuses

S5: 13-common-causes.pdf — 2 Voltage imbalance

S6: 13-common-causes.pdf — 7 Misalignment

S7: 13-common-causes.pdf — 10 Bearing wear

S8: NP_7502_Electric Motor Predictive and Preventive Maintenance Guide.pdf — Moisture and motor insulation

S9: ac-troubleshooting-guide.pdf — Unbalanced load and bearing failure

S10: ac-troubleshooting-guide.pdf — Defective winding

S11: Manual_Udrzba_Montaz_TIMKEN_AJ.pdf — Damaged bearings

S12: Manual_Udrzba_Montaz_TIMKEN_AJ.pdf — Bearing repair

Distinct source-scoped concepts retain distinct IDs, even with matching names.
Arrow numbers refer to the evidence pages that follow. No new connections are inferred.
Colors show the first matching concept role in legend order. Technical approval remains pending.

Knowledge graph — source evidence
Arrow references — qualifiers and conditions belong to the claim.

13-common-causes.pdf · 7 Misalignment
[1] Misalignment → DETECTED_BY → Fluke 830 Laser Shaft Alignment Tool. Qualifier: to diagnose and correct misalignment.
Page 5: Fluke 830 Laser Shaft Alignment Tool to diagnose and correct misalignment

Electric Motor Troubleshooting (Polyphase).pdf · Motor will not start; hums and heats
[2] Input circuit single phased → CAUSES → Motor will not start (Just hums and heats up.). Qualifier: could have been a temporary condition.
Page 6: Cause 3a: Input circuit single phased (No voltage in one or two phases; could have been a temporary condition).

Manual_Udrzba_Montaz_TIMKEN_AJ.pdf · Bearing repair
[3] raised or rough areas → MITIGATED_BY → smoothing flat with grinding and polishing tools. Qualifier: should.
Page 12: Any raised or rough areas should be smoothed flat with grinding and polishing tools.

13-common-causes.pdf · 7 Misalignment
[4] Misalignment → DETECTED_BY → Fluke 810 Vibration Tester. Qualifier: to identify potential misalignment issues.
Page 5: Fluke 810 Vibration Tester to identify potential misalignment issues

NP_7502_Electric Motor Predictive and Preventive Maintenance Guide.pdf · Moisture and motor insulation
[5] excessive moisture → CAUSES → failures in motor leads. Qualifier: causes.
Page 54: Excessive moisture causes failures in motor winding insulation and in motor leads

ac-troubleshooting-guide.pdf · Defective winding
[6] Defective winding → MITIGATED_BY → replace the motor. Qualifier: asserted.
Page 3: For defective winding, replace the motor.

3GZF500730-85 RevH_EN 06-2022_lowres.pdf · Motor fails to start — fuses
[7] Blown fuses → CAUSES → Motor fails to start. Qualifier: asserted.
Page 25: Blown fuses
Page 25: Motor fails to start

3GZF500730-85 RevH_EN 06-2022_lowres.pdf · Motor fails to start — fuses
[8] Motor → HAS_FAILURE_MODE → Motor fails to start. Qualifier: asserted.
Page 25: Motor fails to start

Manual_Udrzba_Montaz_TIMKEN_AJ.pdf · Bearing repair
[9] Small areas of damage on the contact surface of the rolling elements → MITIGATED_BY → grinding out the loose metal. Qualifier: can sometimes.
Page 12: Small areas of damage on bearing races, and on the contact sur- face of the rolling elements, can sometimes be repaired by grind- ing out the loose metal.

13-common-causes.pdf · 10 Bearing wear
[10] failed bearing → HAS_SYMPTOM → lower efficiency. Qualifier: asserted.
Page 6: has lower efficiency

13-common-causes.pdf · 2 Voltage imbalance
[11] Imbalance → CAUSES → excessive current flow in one or more phases. Qualifier: creates.
Page 2: Imbalance creates excessive current flow in one or more phases

3GZF500730-85 RevH_EN 06-2022_lowres.pdf · Motor stalls
[12] Wrong application → CAUSES → Motor stalls. Qualifier: asserted.
Page 25: Wrong application
Page 25: Motor stalls

13-common-causes.pdf · 10 Bearing wear
[13] normal wear → CAUSES → bearing failure. Qualifier: can.
Page 6: Normal wear

Manual_Udrzba_Montaz_TIMKEN_AJ.pdf · Bearing repair
[14] Small areas of damage on bearing races → MITIGATED_BY → grinding out the loose metal. Qualifier: can sometimes.
Page 12: Small areas of damage on bearing races, and on the contact sur- face of the rolling elements, can sometimes be repaired by grind- ing out the loose metal.

13-common-causes.pdf · 10 Bearing wear
[15] shaft misalignment → CAUSES → bearing failure. Qualifier: can.
Page 6: Shaft misalignment

NP_7502_Electric Motor Predictive and Preventive Maintenance Guide.pdf · Moisture and motor insulation
[16] wet winding → DETECTED_BY → insulation resistance testing. Qualifier: can often be found by.
Page 54: Indication of a wet winding can often be found by insulation resistance testing

3GZF500730-85 RevH_EN 06-2022_lowres.pdf · Motor stalls
[17] Overload → MITIGATED_BY → Reduce load.. Qualifier: asserted.
Page 25: Reduce load.

3GZF500730-85 RevH_EN 06-2022_lowres.pdf · Motor stalls
[18] One phase may be open → MITIGATED_BY → Check lines for open phase.. Qualifier: asserted.
Page 25: Check lines for open phase.

Manual_Udrzba_Montaz_TIMKEN_AJ.pdf · Damaged bearings
[19] normal wear in use → CAUSES → damaged bearing. Qualifier: may.
Page 12: normal wear in use

Manual_Udrzba_Montaz_TIMKEN_AJ.pdf · Damaged bearings
[20] improper storage and handling → CAUSES → damaged bearing. Qualifier: may.
Page 12: improper storage and handling

3GZF500730-85 RevH_EN 06-2022_lowres.pdf · Motor stalls
[21] Wrong application → MITIGATED_BY → Change type or size. Consult equipment supplier.. Qualifier: asserted.
Page 25: Change type or size. Consult equipment supplier.

Electric Motor Troubleshooting (Polyphase).pdf · Motor will not start; hums and heats
[22] Motor was single phased → MITIGATED_BY → expert check the motor windings to determine whether the single phasing occurred in the input circuit or in the motor windings. Qualifier: asserted.
Page 6: Cure 3b: Replace the motor. ALWAYS have an expert check the motor windings to determine whether the single phasing occurred in the input circuit or in the motor windings. If in the input circuit, correct the single phase condition.

ac-troubleshooting-guide.pdf · Defective winding
[23] Defective winding → HAS_SYMPTOM → high amps. Qualifier: may be.
Page 3: The amps may also be high.

3GZF500730-85 RevH_EN 06-2022_lowres.pdf · Motor fails to start — fuses
[24] Blown fuses → MITIGATED_BY → Replace fuses with proper type and rating.. Qualifier: asserted.
Page 25: Replace fuses with proper type and rating.

13-common-causes.pdf · 10 Bearing wear
[25] failed bearing → HAS_SYMPTOM → increased drag. Qualifier: asserted.
Page 6: A failed bearing has increased drag

13-common-causes.pdf · 10 Bearing wear
[26] incorrect fit → CAUSES → bearing failure. Qualifier: can.
Page 6: Incorrect fit

ac-troubleshooting-guide.pdf · Defective winding
[27] Defective winding → DETECTED_BY → test the winding for shorted or open circuits. Qualifier: can be detected by.
Page 3: Test the winding for shorted or open circuits.

Electric Motor Troubleshooting (Polyphase).pdf · Motor will not start; hums and heats
[28] Motor was single phased → CAUSES → Motor will not start (Just hums and heats up.). Qualifier: asserted.
Page 6: Cause 3b: Motor was single phased as indicated by heat damage to one or two phases of the motor windings.

Electric Motor Troubleshooting (Polyphase).pdf · Starting: motor controller
[29] motor controller → HAS_FAILURE_MODE → will not operate. Qualifier: asserted.
Page 6: Cause 2a: The motor controller will not operate.

13-common-causes.pdf · 10 Bearing wear
[30] induced shaft voltages → CAUSES → bearing failure. Qualifier: can.
Page 6: Induced shaft voltages

3GZF500730-85 RevH_EN 06-2022_lowres.pdf · Motor stalls
[31] Low voltage → CAUSES → Motor stalls. Qualifier: asserted.
Page 25: Low voltage
Page 25: Motor stalls

3GZF500730-85 RevH_EN 06-2022_lowres.pdf · Motor stalls
[32] Overload → CAUSES → Motor stalls. Qualifier: asserted.
Page 25: Overload

Page 25: Motor stalls

NP_7502_Electric Motor Predictive and Preventive Maintenance Guide.pdf · Moisture and motor insulation
[33] excessive moisture → CAUSES → failures in motor winding insulation. Qualifier: causes.
Page 54: Excessive moisture causes failures in motor winding insulation

13-common-causes.pdf · 10 Bearing wear
[34] heavier load than designed for → CAUSES → bearing failure. Qualifier: can.
Page 6: A heavier load than designed for

NP_7502_Electric Motor Predictive and Preventive Maintenance Guide.pdf · Moisture and motor insulation
[35] excessive moisture → CAUSES → deterioration of motor parts, both electrical and mechanical. Qualifier: can also cause.
Page 54: and deterioration of motor parts, both electrical and mechanical

3GZF500730-85 RevH_EN 06-2022_lowres.pdf · Motor stalls
[36] Low voltage → MITIGATED_BY → Ensure the rating plate voltage is maintained. Check connection.. Qualifier: asserted.
Page 25: Ensure the rating plate voltage is maintained. Check connection.

Manual_Udrzba_Montaz_TIMKEN_AJ.pdf · Bearing repair
[37] Light rust or corrosion → MITIGATED_BY → removal with emery paper (240 – 320 # grit). Qualifier: should.
Page 12: Light rust or corrosion should be removed with emery paper (240 – 320 # grit).

Manual_Udrzba_Montaz_TIMKEN_AJ.pdf · Damaged bearings
[38] damaged bearing → MITIGATED_BY → repair or refurbishment. Qualifier: may.
Page 12: may still be returned to service after repair or refurbishment

13-common-causes.pdf · 2 Voltage imbalance
[39] excessive current flow in one or more phases → CAUSES → increased operating temperatures. Qualifier: increases.
Page 2: that then increases operating temperatures

13-common-causes.pdf · 10 Bearing wear
[40] ineffective bearing sealing → CAUSES → bearing failure. Qualifier: can.
Page 6: Ineffective bearing sealing

NP_7502_Electric Motor Predictive and Preventive Maintenance Guide.pdf · Moisture and motor insulation
[41] excessive moisture → CAUSES → loss of lubrication capability in bearing oil systems. Qualifier: can also cause.
Page 54: It can also cause loss of lubrication capability in bearing oil systems

ac-troubleshooting-guide.pdf · Unbalanced load and bearing failure
[42] unbalanced load → CAUSES → bearings to fail. Qualifier: will also.
Page 3: An unbalanced load will also cause the bearings to fail.

13-common-causes.pdf · 2 Voltage imbalance
[43] increased operating temperatures → LEADS_TO → insulation breakdown. Qualifier: leading to.

Page 2: leading to insulation breakdown

13-common-causes.pdf · 10 Bearing wear
[44] failed bearing → HAS_SYMPTOM → emits more heat. Qualifier: asserted.
Page 6: emits more heat

3GZF500730-85 RevH_EN 06-2022_lowres.pdf · Motor stalls
[45] One phase may be open → CAUSES → Motor stalls. Qualifier: may.
Page 25: One phase may be open
Page 25: Motor stalls

Electric Motor Troubleshooting (Polyphase).pdf · Motor will not start; hums and heats
[46] Input circuit single phased → MITIGATED_BY → qualified technician check for the proper voltage in all three phases. Qualifier: asserted.
Page 6: Cure 3a: Have a qualified technician check for the proper voltage in all three phases. If a single phase condition exists, correct the problem.

13-common-causes.pdf · 10 Bearing wear
[47] inadequate or incorrect lubrication → CAUSES → bearing failure. Qualifier: can.
Page 6: Inadequate or incorrect lubrication